

EXPERIMENT 1: Navigating Local GUI Directories and Managing Files/Folders

1. Aim

To learn and perform basic file and folder management operations using the graphical user interface (GUI) of an operating system, including **Create, Copy, Move, Rename, Delete, Compress (ZIP), and Extract** operations.

2. Objectives

After completing this experiment, students will be able to:

1. Navigate through local directories.
2. Create files and folders.
3. Copy and move files/folders.
4. Rename and delete files/folders.
5. Compress files into a ZIP archive.
6. Extract files from a ZIP archive.
7. Understand the organization of files and folders in a computer.

3. Requirements

- Computer/Laptop
- Windows/Linux operating system
- File Manager/Windows File Explorer
- At least 100 MB free storage

4. Basic Concepts

A **file** is a collection of related data stored with a name and extension.

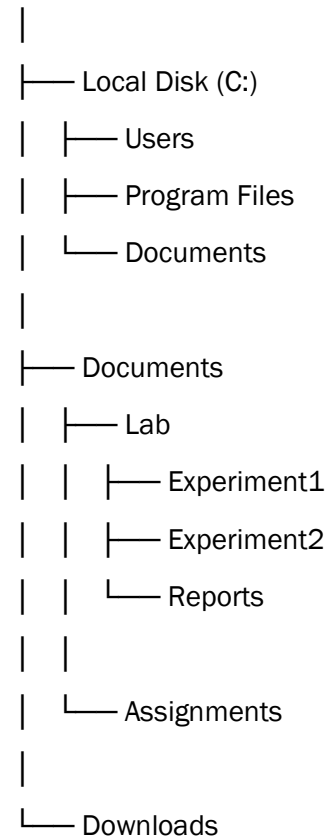
Examples:

- report.docx
- image.jpg
- program.py
- data.xlsx

A **folder/directory** is a container used to organize files and other folders.

5. Directory Structure

Computer



6. Procedure

Step 1: Open File Explorer

1. Start the computer.
2. Click the **File Explorer** icon.
3. Alternatively, press **Windows + E**.
4. The File Explorer window will open.

Step 2: Navigate to Documents

1. Select **Documents** from the left navigation pane.
2. Observe the files and folders available.
3. Navigate into different folders by double-clicking them.
4. Use the **Back** button to return to the previous location.
5. Use the **Address Bar** to view the current directory path.

Example:

This PC → Documents → Lab

Step 3: Create a New Folder

1. Open the **Documents** folder.
2. Right-click on an empty area.
3. Select **New → Folder**.
4. Enter the name:

Computer_Lab

5. Press **Enter**.

Step 4: Create Subfolders

Open Computer_Lab and create the following folders:

Computer_Lab

|

|— Experiment1

|— Documents

|— Images

|— Backup

Step 5: Create a Text File

1. Open the Experiment1 folder.
2. Right-click → **New → Text Document**.
3. Name the file:

Student_Details.txt

4. Open the file.
5. Enter sample information such as:

Name: _____

Roll No.: _____

Course: _____

Experiment: 1

6. Save the file.

Step 6: Copy a File

1. Select Student_Details.txt.
2. Right-click → **Copy**.
3. Open the Documents folder.
4. Right-click → **Paste**.

Alternatively:

Ctrl + C → Navigate to destination → Ctrl + V

Step 7: Move a File

1. Select the copied file.
2. Press **Ctrl + X**.
3. Open the Backup folder.
4. Press **Ctrl + V**.

The file is now moved from its original location to the Backup folder.

Step 8: Rename a File

1. Right-click Student_Details.txt.
2. Select **Rename**.
3. Change the name to:

Student_Record.txt

4. Press **Enter**.

Shortcut:

Select file → F2 → Type new name → Enter

Step 9: Delete a File

1. Select the file.
2. Press the **Delete** key.
3. Confirm the operation if prompted.
4. The file will normally be moved to the **Recycle Bin**.

Step 10: Restore a Deleted File

1. Open **Recycle Bin**.
2. Locate the deleted file.
3. Right-click the file.
4. Select **Restore**.
5. The file will return to its original location.

Step 11: Compress Files into ZIP

1. Create a folder named Lab_Files.
2. Place two or more files inside it.
3. Right-click the folder.
4. Select **Compress to ZIP file** or **Send to → Compressed (zipped) folder**, depending on the Windows version.

5. A ZIP file will be created.

Example:

Lab_Files



Lab_Files.zip

Step 12: Extract ZIP File

1. Right-click Lab_Files.zip.
2. Select **Extract All**.
3. Select the destination folder.
4. Click **Extract**.
5. Verify that the files have been restored.

7. Useful Keyboard Shortcuts

Operation	Shortcut
Open File Explorer	Windows + E
Copy	Ctrl + C
Cut/Move	Ctrl + X
Paste	Ctrl + V
Rename	F2
Delete	Delete
Undo	Ctrl + Z
Select All	Ctrl + A
Search	Ctrl + F
Create new folder	Ctrl + Shift + N

8. Result

The student successfully navigated local directories and performed **create, copy, move, rename, delete, ZIP compression, and extraction** operations.

EXPERIMENT 2: Extracting Local Hardware Configuration

1. Aim

To identify and record the basic hardware configuration of a computer, including **CPU, RAM, storage capacity, operating system, and other system information.**

2. Objectives

Students will learn how to:

1. Identify the processor/CPU.
2. Check installed RAM.
3. Check storage capacity.
4. Identify the operating system.
5. View detailed hardware information.
6. Record system configuration for laboratory documentation.

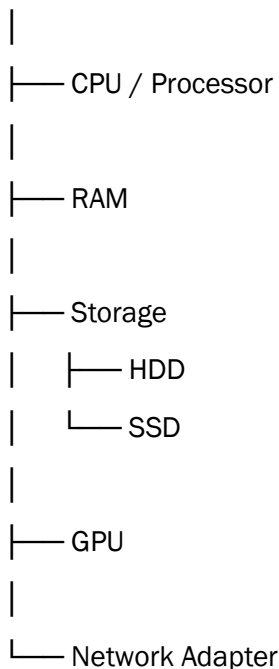
3. Requirements

- Computer/Laptop
- Windows operating system

4. Hardware Configuration

The major hardware components to be examined are:

Computer System



5. Procedure

Part A: Check CPU, RAM and Operating System

Step 1 Press: Windows + I

to open **Settings**.

Step 2 Select: System → About

Step 3

Observe the following information:

- Device name
- Processor
- Installed RAM
- System type
- Windows edition
- Windows version

Example:

Processor: Intel Core i5

Installed RAM: 8 GB

System Type: 64-bit Operating System

Operating System: Windows 11

Part B: Check CPU and RAM Using Task Manager

Step 1

Press: Ctrl + Shift + Esc

Step 2

Task Manager will open.

Step 3

Select the **Performance** tab.

Step 4

Click **CPU**.

Record:

- CPU model
- CPU utilization
- Speed
- Number of cores
- Number of logical processors

Step 5

Click **Memory**.

Record:

- Total RAM
- RAM currently in use
- Available RAM
- Memory speed

Example:

CPU: Intel Core i5

Cores: 4

Logical Processors: 8

RAM: 8 GB

Memory Speed: 3200 MHz

Part C: Check Storage

Step 1

Open:

File Explorer → This PC

Step 2

Observe available drives.

Example:

Local Disk (C:)

Capacity: 512 GB

Free Space: 320 GB

Local Disk (D:)

Capacity: 1 TB

Free Space: 750 GB

Step 3

For more information:

1. Right-click the drive.
2. Select **Properties**.
3. Observe:
 - Used space

- Free space
- Capacity
- File system

Part D: Check Storage Type

1. Open **Task Manager**.
2. Select **Performance**.
3. Select **Disk 0**.
4. Observe whether the drive is an **SSD or HDD**, where displayed.

Part E: Device Manager

Step 1

Right-click the **Start** button.

Step 2

Select:

Device Manager

Step 3

Observe categories such as:

Processors

Display adapters

Network adapters

Disk drives

Keyboards

Mice

Sound controllers

Step 4

Expand each category and record relevant hardware.

6. Hardware Configuration Table

Students should prepare a table similar to the following:

Component	Details
Operating System	_____
Processor	_____
CPU Speed	_____

Component	Details
Number of Cores	_____
RAM	_____
RAM Speed	_____
Storage	_____
Storage Type	SSD/HDD
GPU	_____
Network Adapter	_____
System Type	64-bit/32-bit

7. Result

The hardware and operating-system configuration of the computer was successfully identified and recorded.

EXPERIMENT 3: Local System Administration – File Paths, Shortcuts and Directory Organization

1. Aim

To perform basic local system administration tasks involving **file paths, shortcuts, directory organization, and optimization of local data workflows.**

2. Objectives

Students will learn:

1. Absolute and relative file paths.
2. Creating shortcuts.
3. Organizing files systematically.
4. Searching for files.
5. Managing frequently accessed folders.
6. Designing an efficient directory structure.

3. File Path Concepts

A **file path** specifies the location of a file or folder.

Example of an absolute Windows path:

C:\Users\Student\Documents\Computer_Lab\Experiment1

A simplified directory structure:

C:

```
└─ Users
   └─ Student
      └─ Documents
         └─ Computer_Lab
            ├── Experiment1
            ├── Experiment2
            └─ Reports
```

4. Procedure

Step 1: Identify the Current Path

1. Open File Explorer.
2. Navigate to a folder.
3. Click the address bar.
4. Observe the complete path.

Example:

C:\Users\Student\Documents\Computer_Lab

Step 2: Create an Organized Directory Structure

Create:

GPB_Lab

```
|
|— Lab_01
|— Lab_02
|— Assignments
|— Notes
|— Images
|— Reports
|— Backup
```

Step 3: Organize Files

Place files according to their purpose.

For example:

Notes → Lecture notes

Images → Screenshots

Reports → Lab reports

Backup → Backup copies

Assignments → Assignment files

Avoid keeping all files in the Desktop or Downloads folder.

Step 4: Create a Shortcut

1. Locate the frequently used AI_Lab folder.
2. Right-click the folder.
3. Select **Show more options → Send to → Desktop (create shortcut)**, depending on Windows version.
4. A shortcut will appear on the Desktop.

The shortcut provides quick access without moving the original folder.

Step 5: Rename the Shortcut

Rename it: AI Lab

Step 6: Test the Shortcut

1. Double-click the shortcut.
2. Verify that the original folder opens.

3. Do not delete the original folder while testing.

Step 7: Search for Files

1. Open File Explorer.
2. Navigate to a drive/folder.
3. Use the search box.
4. Search for: *.pd for *.docx

This helps locate files according to their extensions.

Step 8: Sort Files

In File Explorer:

1. Open a folder.
2. Select **Sort**.
3. Sort by:
 - Name
 - Date modified
 - Type
 - Size

This improves file management.

5. Data Workflow Optimization

A good local data workflow can be represented as:

Collect Data



Create Folder



Organize Files



Rename Files Properly



Process Data



Save Results



Create Backup

6. Recommended File Naming Convention

Use meaningful names.

Poor naming:

file1.docx

newfile.docx

finalfinal.docx

abc.pdf

Better naming:

Experiment_01_File_Management.docx

Experiment_02_Hardware_Configuration.docx

Student_Record_2026.pdf

Lab_Report_01.docx

7. Result

The student successfully created an organized directory structure, identified file paths, created shortcuts, searched files, and implemented an efficient local data-management workflow.

EXPERIMENT 4: Collaborative Cloud Directory Configuration Using Google Drive

1. Aim

To create and configure a collaborative cloud directory using **Google Drive**, upload files in multiple formats, share files with other users, and configure appropriate security permissions.

2. Objectives

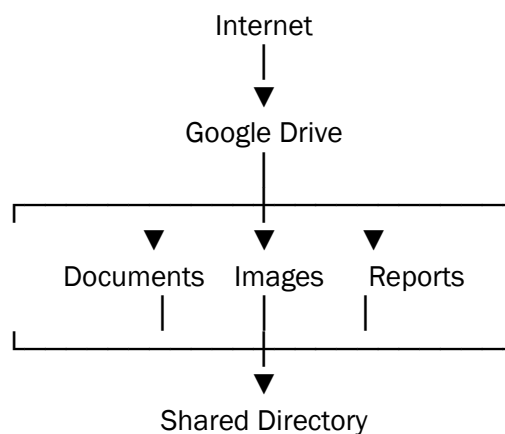
Students will learn:

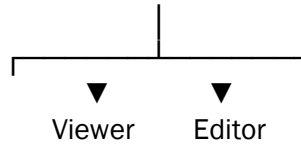
1. Create cloud folders.
2. Upload files.
3. Upload different file formats.
4. Create and share folders.
5. Configure Viewer and Editor permissions.
6. Share screenshots/documentation.
7. Apply safe cloud-storage practices.

3. Requirements

- Computer/Laptop
- Internet connection
- Google account
- Web browser
- Files in different formats such as:
 - .docx
 - .pdf
 - .jpg/.png
 - .xlsx
 - .txt

4. Cloud Storage Concept





5. Procedure

Step 1: Open Google Drive

1. Open a web browser.
2. Sign in to your Google account.
3. Open Google Drive.
4. Verify that you are using the correct account.

Step 2: Create a Cloud Folder

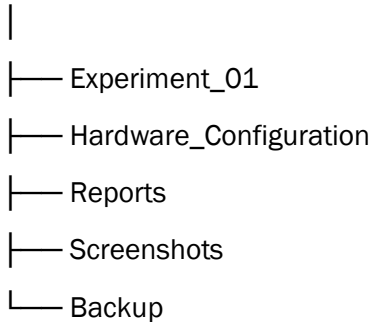
1. Click **New**.
2. Select **New folder**.
3. Enter:
AI_Computer_Lab
4. Click **Create**.

Step 3: Create Subfolders

Open AI_Computer_Lab.

Create:

AI_Computer_Lab



Step 4: Upload a Document

1. Open Experiment_01.
2. Click **New**.
3. Select **File upload**.
4. Select a .docx or .pdf file.
5. Wait until the upload is complete.
6. Verify that the file appears in the folder.

Step 5: Upload an Image

1. Open the Screenshots folder.
2. Click **New → File upload**.
3. Select a .jpg or .png screenshot.
4. Wait for upload completion.
5. Verify the uploaded image.

Step 6: Upload Multiple File Formats

Upload at least:

Lab_Report.docx

Experiment.pdf

Screenshot.png

Data.xlsx

Readme.txt

This demonstrates multi-format cloud data management.

Step 7: Create a Google Document

1. Click **New**.
2. Select **Google Docs**.
3. Enter a title such as:

Computer Lab Record

4. Enter basic experiment information.
5. The document will automatically be saved in Google Drive.